



KE: Introduction

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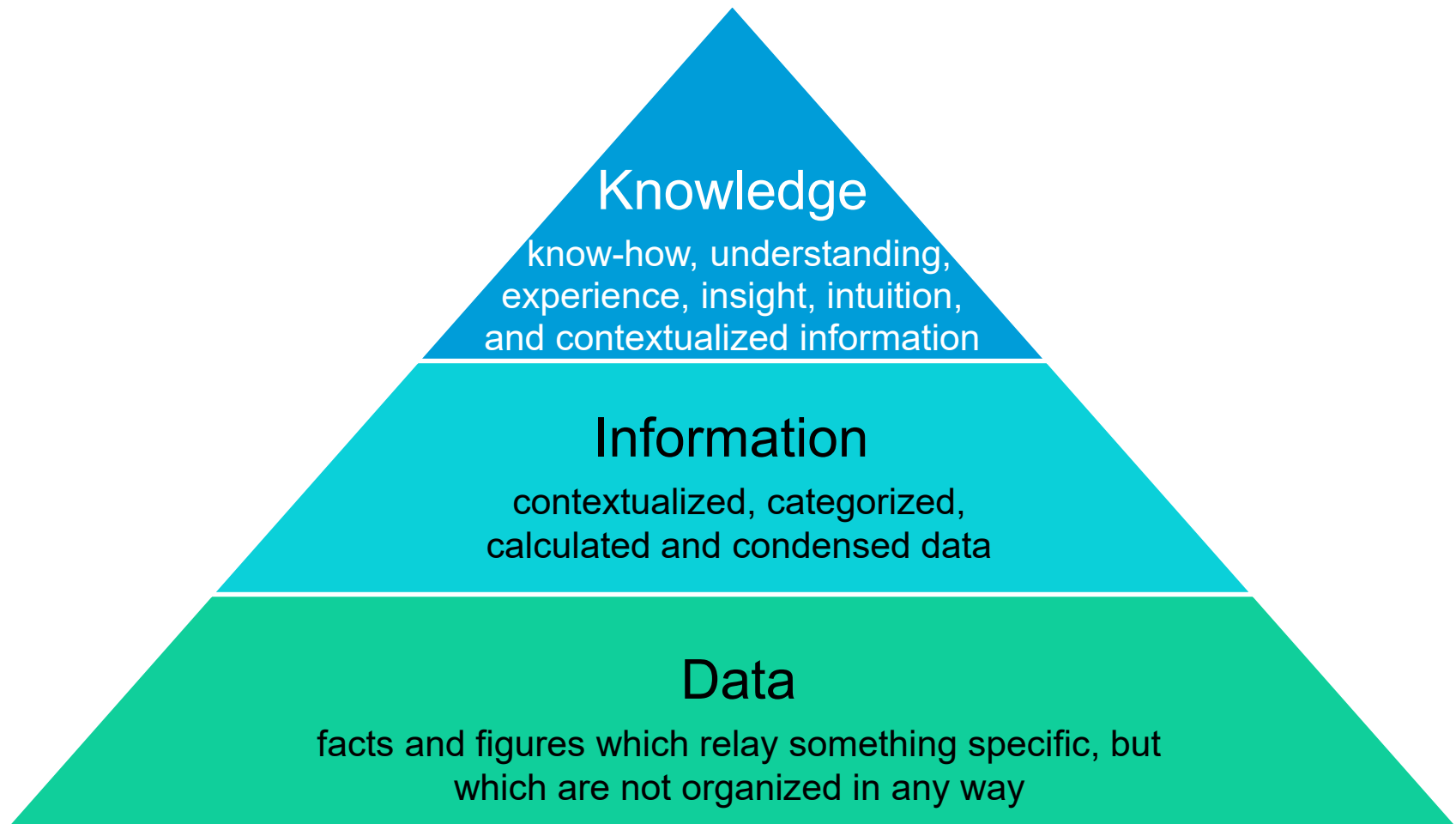
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Knowledge vs Information vs Data



Data

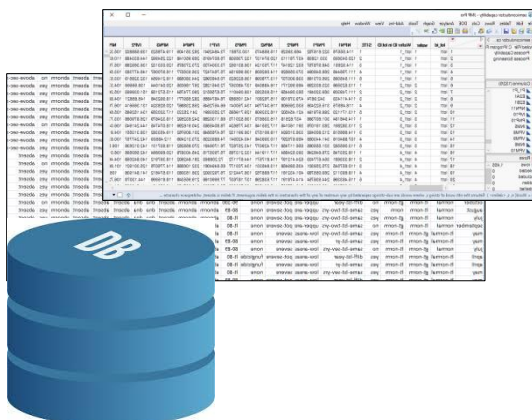
- A "given" or a fact; number; picture
- Represents something in real world
- Raw materials in production of information

Information

- Data that have meaning in context
- Data related
- Data after manipulation

Knowledge

- Familiarity, awareness and understanding of someone or something
- Acquired through experience or learning



Detecting fraud

Package offering based on customer characteristics

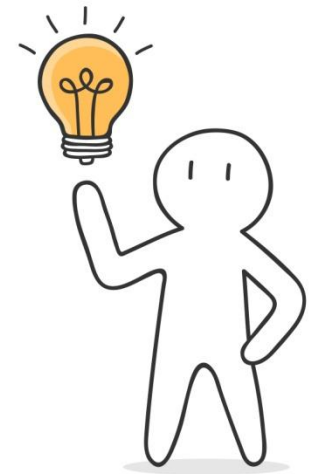
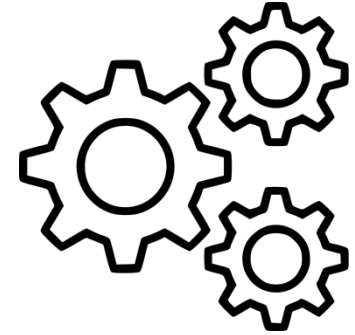
Suggestion based on certain set of conditions

How to fix a car

Diagnosis

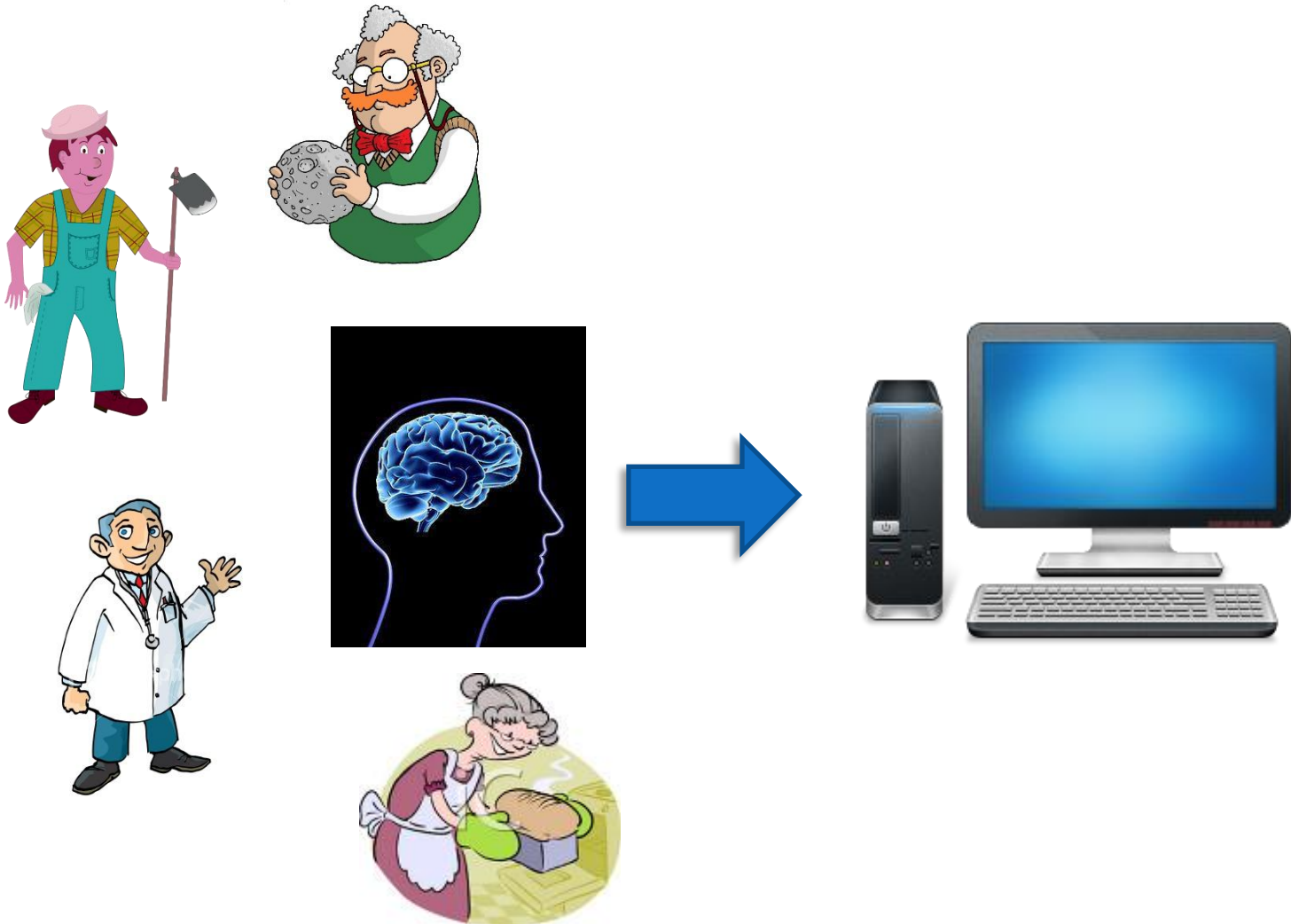
How to treat a patient with specific symptoms

Deep NN
NN Prob. DT Markov
Rules Math model





Human Knowledge in Computer-Understandable Form





What is Knowledge?

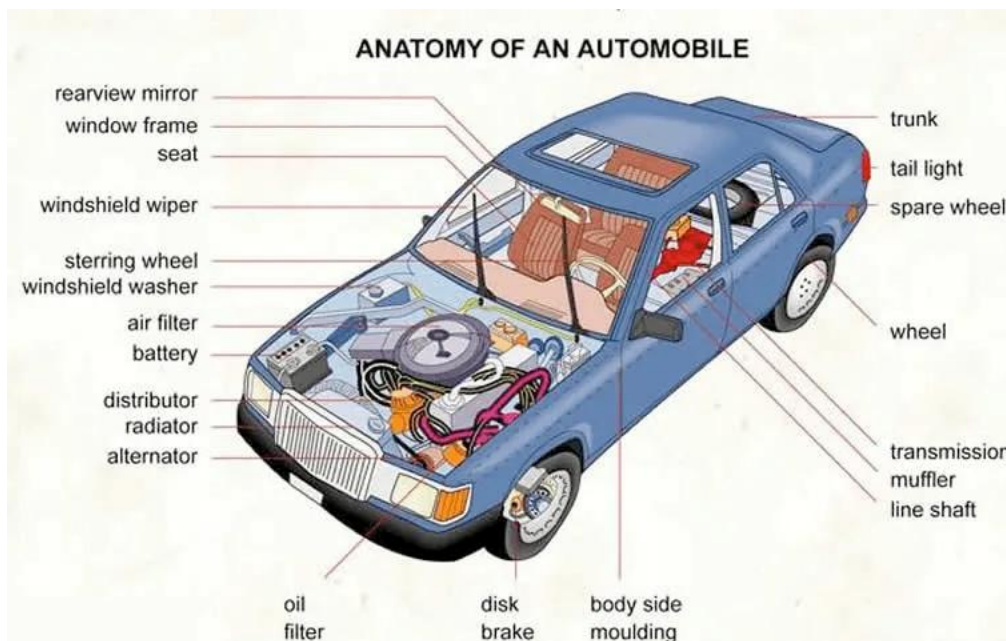
- An awareness or understanding of something, including
 - Descriptive knowledge (facts): Theory (Knowing what)
 - Procedural knowledge (skills): Decision-making/Execution (Knowing how)
 - Acquaintance knowledge (objects): Familiarity/Perception



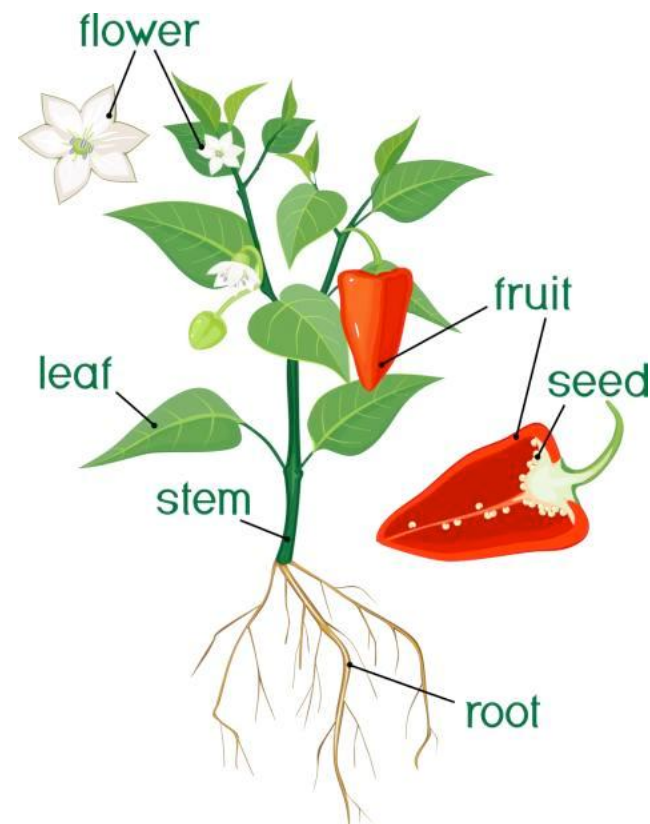
Knowledge

- Explicit knowledge
 - This type of knowledge is formalized and codified, and is sometimes referred to as *know-what*
 - Fairly easy to identify, store, and retrieve
 - Most easily handled by KMS
- Tacit knowledge
 - Sometimes referred to as *know-how*
 - Refers to intuitive, hard to define knowledge that is largely experience based.
 - Tacit knowledge is often context dependent and personal in nature

Knowing what



- What compositions are the things made of?
- What function of each part and whole?
- How it relates to other?



cf: "Representing and Managing Knowledge with Ontology: Ontology Engineering",
Dr. Taneth Ruangrajitpakorn



Knowledge Engineering

- Knowledge engineering is a field within AI
- Focus on developing systems that capture, represent, and use human knowledge to solve complex problems.
- Knowledge engineering involves translating the expertise of human specialists into computational models that can be used by computers to perform tasks typically requiring human intelligence.

It's about making computers "think" like experts in a specific field.



Key steps of Knowledge Engineering

- **Knowledge Acquisition**
 - Gathering and documenting knowledge from human experts, including facts, rules, and problem-solving strategies.
- **Knowledge Representation**
 - Converting the acquired knowledge into a formal structure that a computer can understand and process.
- **Inference and Reasoning**
 - Developing mechanisms for the system to use the knowledge to draw conclusions, solve problems, and make decisions.
- **Explanation and Justification**
 - Enabling the system to explain its reasoning and the basis for its conclusions.
 - This is crucial for building trust and understanding the system's behavior

In this class ...

Manipulation of knowledge is a heart of what intelligent beings do

Design structure of knowledge

Represent knowledge using symbolic structures

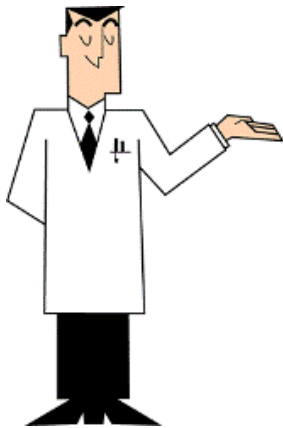
Perform formalized versions of reasoning



- In this class, we will learn how to formalize and represent knowledge using a representation called “**Ontology**”

Reasoning / Inference

- It is not always possible, and it's not desirable, to store all knowledge as explicit facts.
- We want to be able to infer new knowledge from explicit facts.

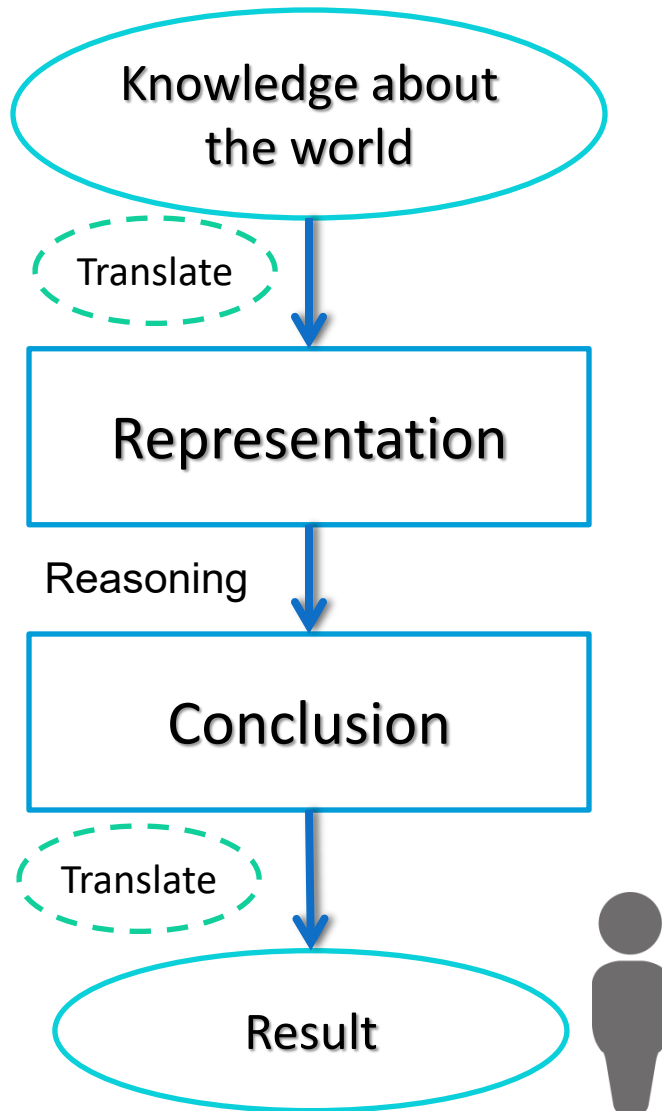


You want to answer the kin relationship between pair of people

We DON'T want to store a every possible kin relations between any pair of people.

Target: We want to be able to infer such relations from a basic set of relations/information

- However, more complex inferences are likely to take longer.



To translate confidently, we need well-defined syntax and semantics

For example:

Does *Like(Jane, Pizza)* mean Jane likes all pizza, or that she just likes the one that she has eaten.



Application with knowledge representation

- Expert system, diagnosis system, recommendation system
 - For example: [EasyDiagnosis](#), [Isabel Symptom Checker](#), etc.
- Representing data with complex relationships
 - For example: [GRAMENE](#), [MISO](#)



In this course

Representing knowledge with Ontology